Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/24/2006
PI Name:	Halloran, Bernard Ph.D.		
Project Title:	Space Flight and Bone Metabolism: Age Effects and Development of an Animal Model for Adult Human Bone Loss		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	bernard.halloran@ucsf.edu	**Fax:**	FY 415-750-6929
PI Organization Type:	UNIVERSITY	**Phone:**	415-750-6928
Organization Name:	University of California, San Francisco		
PI Address 1:	VA Medical Center		
PI Address 2:	4150 Clement Street		
PI Web Page:			
City:	San Francisco	**State:**	CA
Zip Code:	94121	**Congressional District:**	8
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-02
Start Date:	10/01/2000	**End Date:**	07/31/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC2-1158		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Devise a means of remotely labeling bones to permit the measurement of bone formation during space flight.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:	As part of our flight experiment we proposed to fluorochrome label bones in flight using osmotic minipumps. Previous studies have not been able to label bones during flight and thus measurements of bone formation during different periods of the flight has not been possible.		

Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/25/2014)